Question

Given: At 693 K, the initial amounts of Xe and F_2 in the container are 2 mol and 1 mol, respectively.

Reaction equations:

$$Xe(g) + F_2(g) \rightleftharpoons XeF_2(g) \quad K_{p1} = 360.3$$

$$Xe(g) + 2F_2(g) \rightleftharpoons XeF_4(g) \quad K_{p2} = 1978.8$$

$$Xe(g) + 3F_2(g) \rightleftharpoons XeF_6(g) \quad K_{p3} = 35.99$$

The volume of the container is variable, and the total pressure is maintained at 1 bar. For simplicity and convenience in calculations, the following symbolic conventions must be used. In the equilibrium expressions, all partial pressure terms are divided by the standard partial pressure by default, and the units in the problem are all in bar.

Total system pressure	**Initial partial pressure of** $Xe(g)$	**Initial partial pressure of** $F_2(g)$	**Equilibrium partial pressure of** $Xe(g)$	**Equilibrium partial pressure of** $F_2(g)$	**Equilibrium partial pressure of** $XeF_2(g)$	**Equilibrium partial pressure of** $XeF_4(g)$	**Equilibrium partial pressure of** $XeF_6(g)$
p_1	x_0	x_1	y_0	y_1	y_2	y_3	y_4

Then $\frac{y_4}{y_0+100y_1} =$:

A. 4×10^{-7}

B. 8×10^{-7}

C. 2×10^{-6}

D. 1×10^{-7}

E. 8×10^{-6}

F. 4×10^{-6}

G. 8×10^{-8}

H. None of the above options are correct

Answer

Correct Answer: A

Detailed Explanation

From the quantitative relationship:

$$y_0 + y_1 + y_2 + y_3 + y_4 = 1 \quad (1)$$

CHECKPOINT 1 PTS

$y_0 + y_1 + y_2 + y_3 + y_4 = 1 \quad (1)$

From the molar relationship:

$$x_0 = 2x_1$$

From the atomic conservation relationship:

$$y_0 + y_2 + y_3 + y_4 = 2y_1 + 2y_2 + 4y_3 + 6y_4$$

Combining with (1) yields:

$$2y_2 + 4y_3 + 6y_4 = 1 - 3y_1 \quad (2)$$

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$2y_2 + 4y_3 + 6y_4 = 1 - 3y_1 \quad (2)$

Equilibrium constant expressions:

$$y_2 = K_{p1} y_0 y_1 \quad (3)$$

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$y_2 = K_{p1} y_0 y_1 \quad (3)$

$$y_3 = K_{p2} y_0 y_1^2 \quad (4)$$

CHECKPOINT 1 PTS

$y_3 = K_{p2} y_0 y_1^2 \quad (4)$

$$y_4 = K_{p3} y_0 y_1^3 \quad (5)$$

CHECKPOINT 1 PTS

$y_4 = K_{p3} y_0 y_1^3 \quad (5)$

From (1), (3), (4), and (5), y_0 can be expressed as a function of y_1. Substituting into (2) yields a univariate equation in y_1, ultimately solving to:

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$y_1 = 2.633 \times 10^{-3}$ bar

Substituting back into the equations yields:

$$y_0 = 0.5082 \text{ bar}$$

CHECKPOINT 1 PTS

$y_0 = 0.5082$ bar

$$y_2 = 0.482 \text{ bar}$$

CHECKPOINT 1 PTS

$y_2 = 0.482$ bar

$$y_3 = 6.966 \times 10^{-3} \text{ bar}$$

CHECKPOINT 1 PTS

$y_3 = 6.966 \times 10^{-3}$ bar

$$y_4 = 3.255 \times 10^{-7} \text{ bar}$$

CHECKPOINT 1 PTS

$y_4 = 3.255 \times 10^{-7}$ bar

Substituting and calculating gives $\frac{y_4}{y_0+100y_1} = 4.2 \times 10^{-7}$

CHECKPOINT 1 PTS

$\frac{y_4}{y_0+100y_1} = 4.2 \times 10^{-7}$ Option A is correct